TIRA E-COMMERCE DATABASE MANAGEMENT SYSTEM

SQL Query Implementation For E-Commerce Database

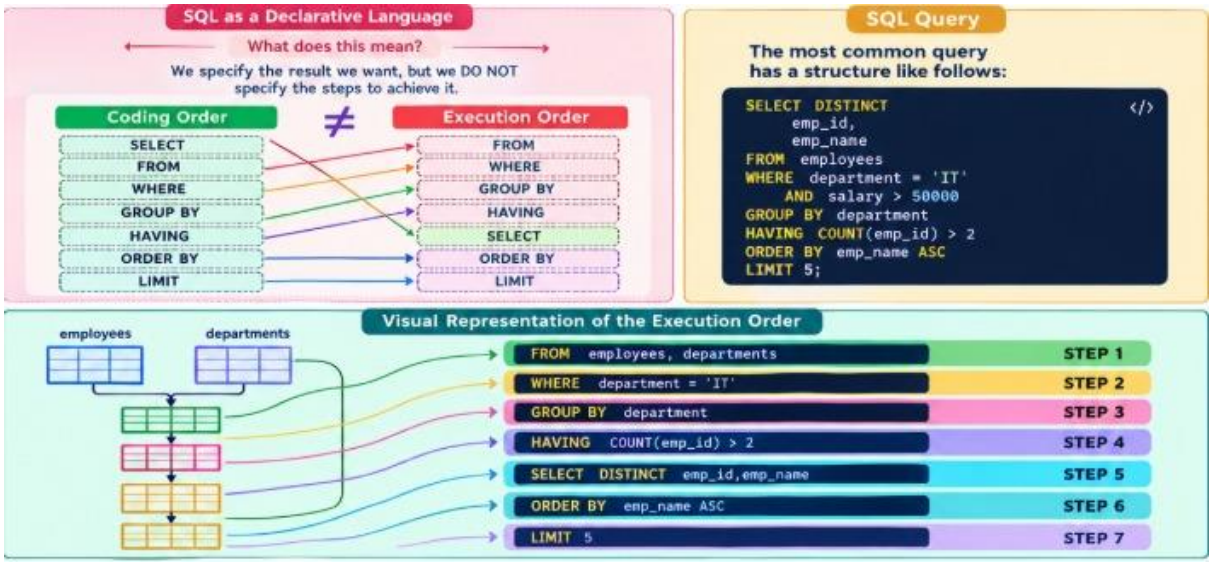
Objectives

- To manage customers and beauty products efficiently.
- To maintain product, brand, and category details.
- To manage customer orders and payments.
- To track product stock and availability.
- To manage customer reviews and ratings.
- To generate reports using SQL queries.
- To ensure data accuracy and consistency.

1.Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all products

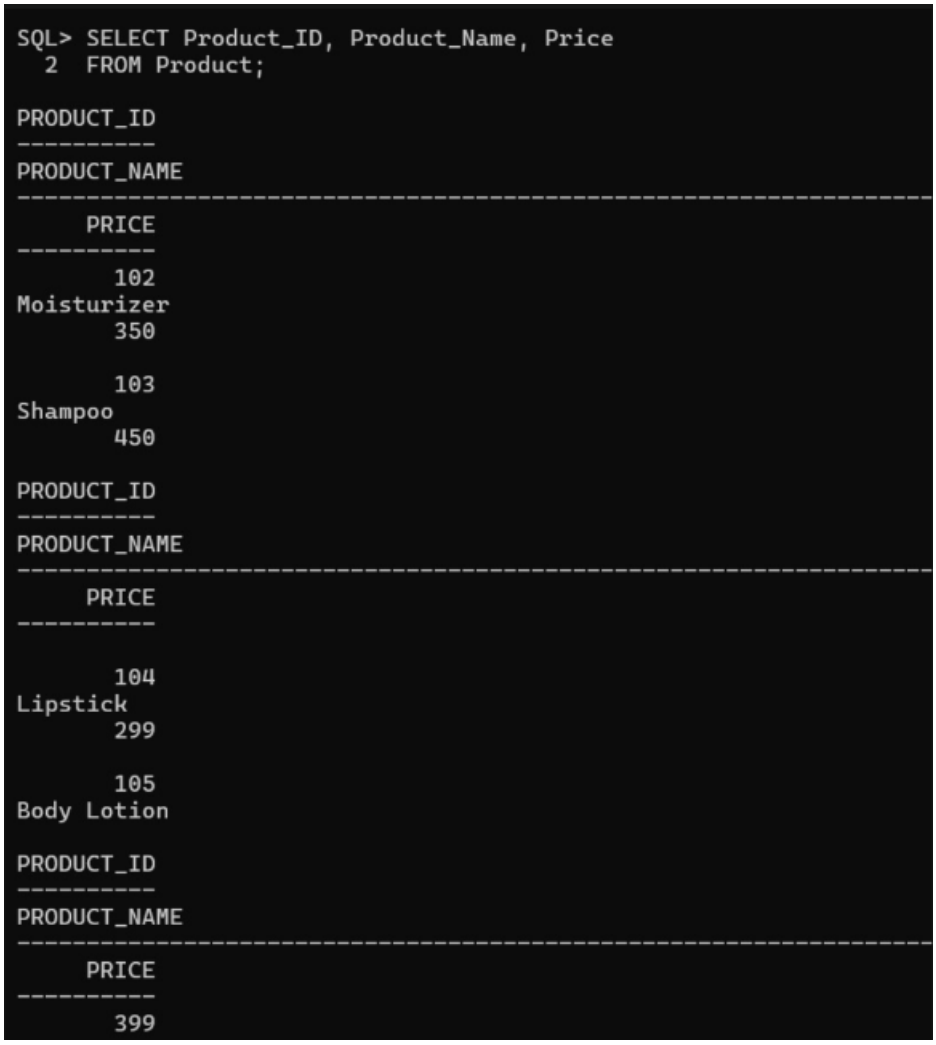
```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ---------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
       102
Moisturizer
       202          2        350

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ---------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
Cream
Dry                                                15-AUG-27         40


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ---------- ----------
SHADE
```

Display selected product details



1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

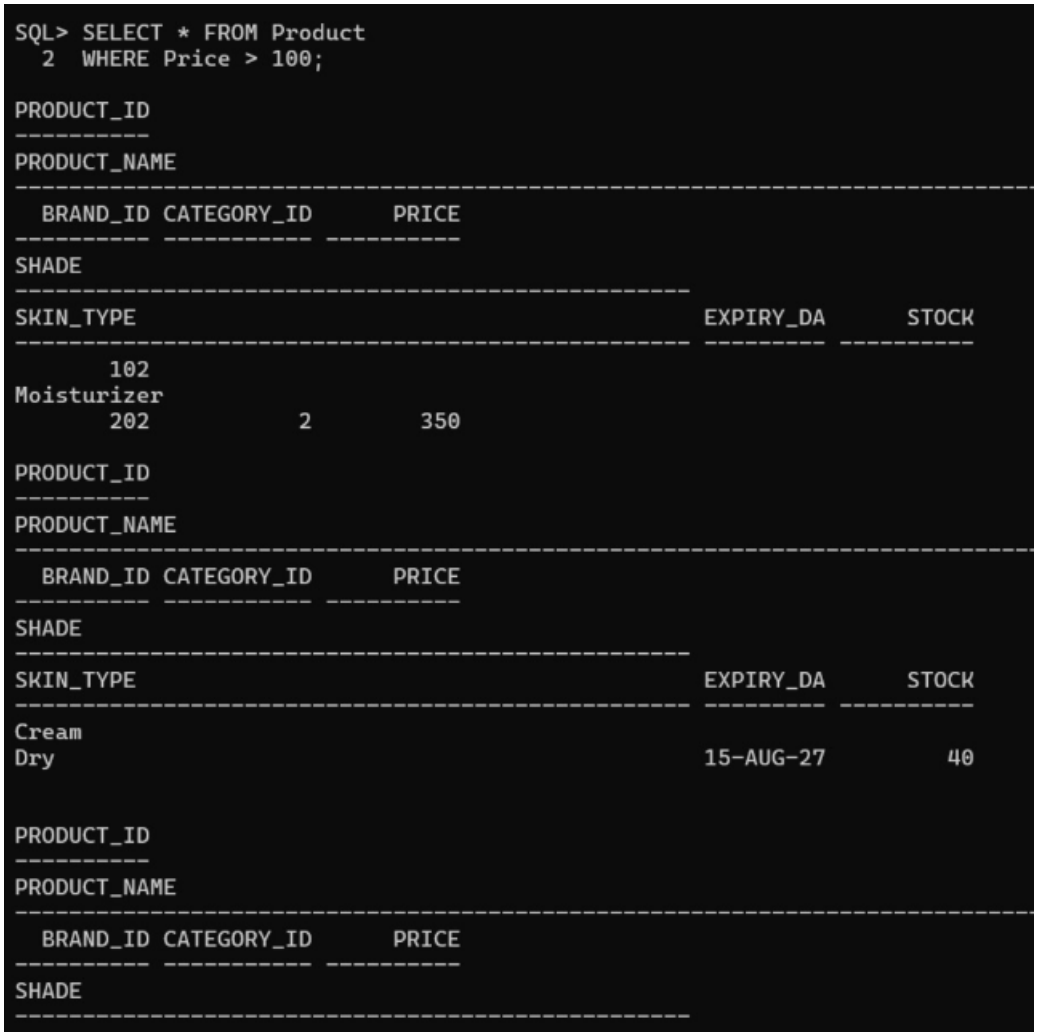
```
SQL> SELECT * FROM Product
  2  WHERE Price < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ---------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
       102
Moisturizer
       202           2        350

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ---------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
Cream
Dry                                                15-AUG-27         40


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ---------- ----------
SHADE
```

Find products with price greater than 100



1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

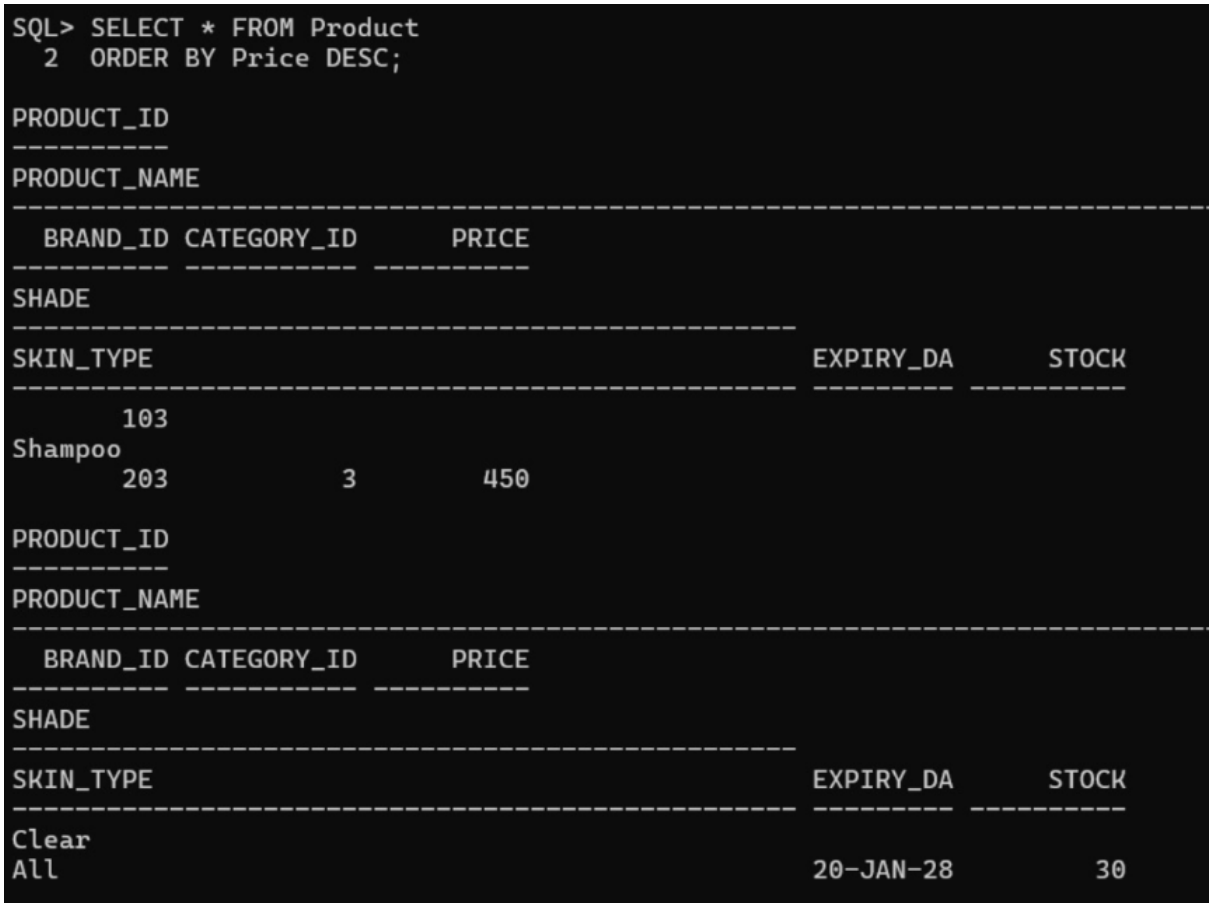
```
SQL> SELECT * FROM Product
  2  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ----------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
       101
Face Wash
       201           1        250

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ----------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
White
Oily                                               10-MAY-27         50


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ----------- ----------
SHADE
--------------------------------------------------
```

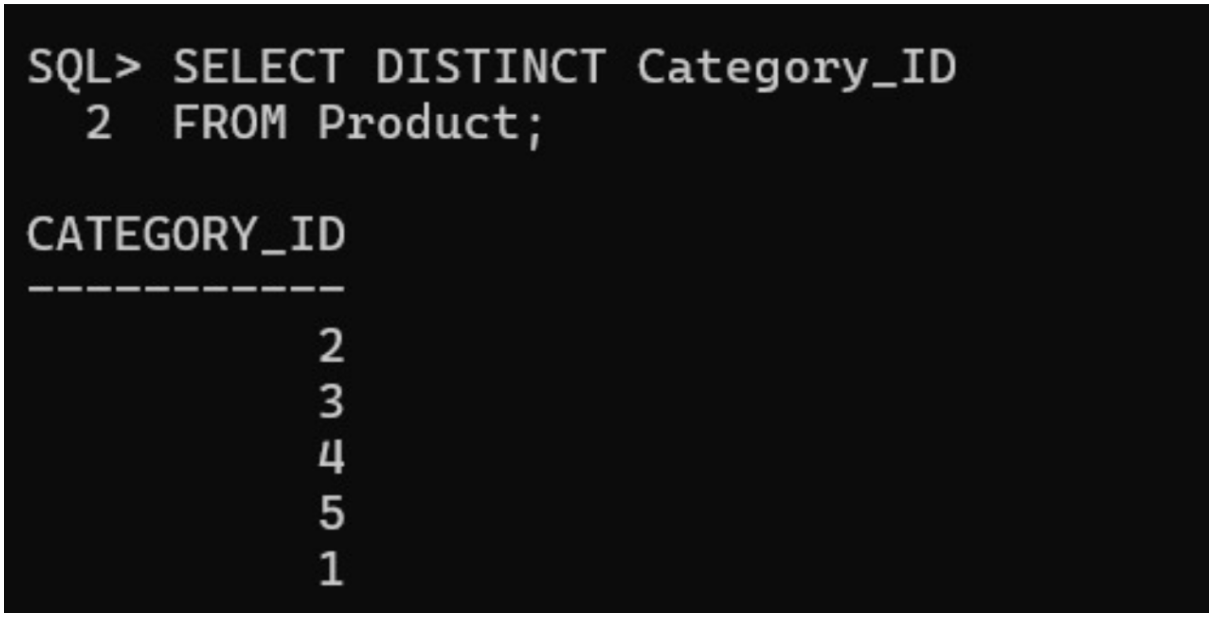
Products from highest to lowest price



1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories



2. Search Products

2.1 Search Products Based on Price

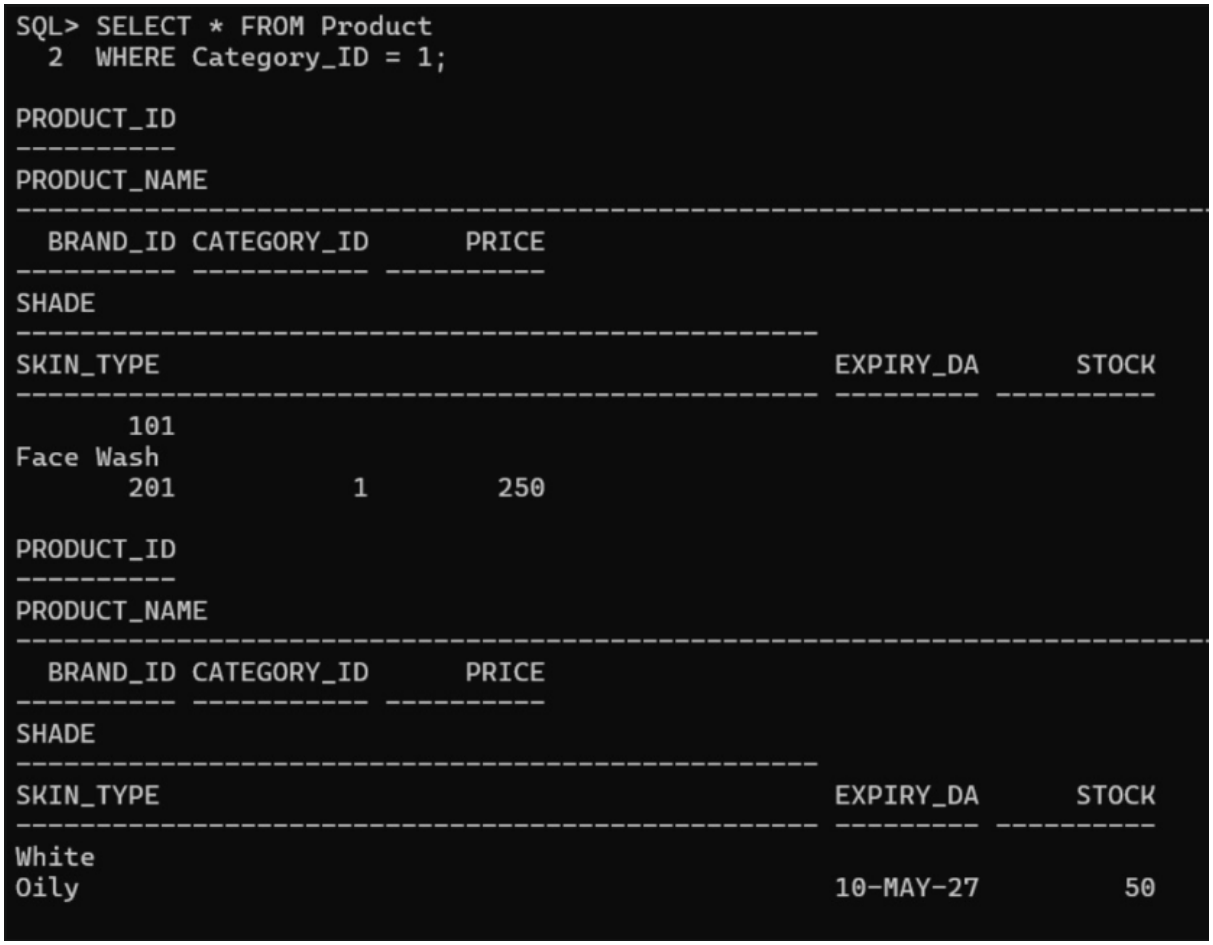
```
SQL> SELECT * FROM Product
  2  WHERE Price BETWEEN 100 AND 400;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ----------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
       102
Moisturizer
       202           2        350

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ----------- ----------
SHADE
--------------------------------------------------
SKIN_TYPE                                          EXPIRY_DA      STOCK
-------------------------------------------------- --------- ----------
Cream
Dry                                                15-AUG-27         40


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE
---------- ----------- ----------
```

2.2 Search Products Based on Category



3. Retrieve Customer Information

Display all customers

```
SQL> SELECT*
  2  FROM CUSTOMER;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE           DATE_OF_B GENDER
--------------- --------- ----------
ADDRESS
--------------------------------------------------------------------
        101 Harini
N
harini101@gmail.com

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE           DATE_OF_B GENDER
--------------- --------- ----------
ADDRESS
--------------------------------------------------------------------
9876543210      15-JUN-05 Female
Chennai


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE           DATE_OF_B GENDER
--------------- --------- ----------
ADDRESS
--------------------------------------------------------------------
        102 Priya
S
priya102@gmail.com
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 101;

CUSTOMER_ID FIRST_NAME
----------- ----------------------------------------
LAST_NAME
----------------------------------------
EMAIL
----------------------------------------
PHONE           DATE_OF_B GENDER
--------------- --------- ----------
ADDRESS
----------------------------------------
        101 Harini
N
harini101@gmail.com

CUSTOMER_ID FIRST_NAME
----------- ----------------------------------------
LAST_NAME
----------------------------------------
EMAIL
----------------------------------------
PHONE           DATE_OF_B GENDER
--------------- --------- ----------
ADDRESS
----------------------------------------
9876543210      15-JUN-05 Female
Chennai
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price < 2000
  4  AND Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       102
Moisturizer
       350
```

4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Category_ID = 1
  4  OR Category_ID = 3;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       103
Shampoo
       450

       101
Face Wash
       250

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
```

4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Category_ID IN (1, 2, 3);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       102
Moisturizer
       350

       103
Shampoo
       450

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

       101
Face Wash
       250
```

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 100 AND 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       104
Lipstick
       299

       101
Face Wash
       250

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       103
Shampoo
       450

       105
Body Lotion
       399

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

       102
Moisturizer
       350

       104
Lipstick

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       299

       101
Face Wash
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      o.Order_ID,
  6      o.Order_Date,
  7      o.Total_Amount
  8  FROM Customer c
  9  JOIN Orders o
 10      ON c.Customer_ID = o.Customer_ID
 11  ORDER BY o.Order_Date DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Harini
N                                                         201 23-OCT-26
        2500

        103 Anitha
R                                                         203 23-OCT-26
        3200

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

        102 Priya
S                                                         202 23-OCT-26
        1500
```

5.3 Product Order Report

```
SQL> SELECT
  2      od.Order_ID,
  3      od.Product_ID,
  4      od.Quantity,
  5      od.Unit_Price,
  6      od.Subtotal
  7  FROM Order_Details od
  8  ORDER BY od.Subtotal DESC;

  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ---------- ----------
       203        104          2       1600       3200
       201        101          3       1000       3000
       202        103          2        900       1800
       201        102          1        500        500
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.First_Name
  9  ORDER BY Total_Orders DESC;

CUSTOMER_ID FIRST_NAME                                         TOTAL_ORDERS
----------- -------------------------------------------------- ------------
        101 Harini                                                        1
        103 Anitha                                                        1
        102 Priya                                                         1
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.First_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID FIRST_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        103 Anitha                                                      3200
        101 Harini                                                      2500
        102 Priya                                                       1500
```